METHOD STATEMENT FOR DISMANTLE OF EXISTING MASTIC TANK WITH LINER & ERECTION OF NEW LINER TANK. Rev 0.

Client:	Richmond Fire Engineers
Contract No:	24-T4265
Site Address:	GXO – Decathlon Wellingborough 39 - 44 Davy Close Park Farm Wellingborough Northampton NN8 6XX

Approx. Dismantle Start Date:	20/5/2024
Approx. Build Start Date:	10/6/2024

THE METHOD OF CONSTRUCTION SHOULD FOLLOW THIS PROCEDURE. IT IS AT THE SITE CREW FOREMAN'S DISCRESSION TO CHANGE THE ORDER OF THE BUILD AND THE METHOD OF ANY TASK WERE DEEMED NECESSARY. AFTER CONSULTATION AND AGREEMENT WITH CLIENT / SITE REPRESENTATIVE AND CST HEALTH AND SAFETY REPRESENTATIVE. ANY SUCH AMENDMENT SHOULD BE DETAILED ON THE ATTACHED AMENDMENT SHEET AND SIGNED BY BOTH PARTIES.

1. Scope of Work

Dismantle of existing mastic (with liner) cylindrical water tank - 9 Round / 7 High.
Installation of new PVC liner tank model 2038 LPCB Ver 2.0 - 8 Round / 9½ High.

2. CST Project Personnel

Health and Safety Representative:	Rick Woffinden	Tel: 01773 835321
Service Manager:	Ian Sneddon	Tel: 07587 132630

3. CST Site Personnel

Tank Engineers: 2-4 x Tank Engineers (Names TBC)
Certificates Held: CSCS Card, Confined Space (details of qualifications available with the site crew at the site induction)

4. Delivery (on the day of the tank build start)

Tank parts will be delivered utilising a vehicle with lifting Hi-ab.
Parts will be off loaded and stored adjacent to the Tank bases. (CST client to provide suitable storage area)

1) The lorry will deliver the complete tank kit. The driver will be suitably certified.
2) Access to the laydown area in the adjacent car park must be clear and, if necessary, cordoned off to ensure that the public / unauthorised people cannot gain access whilst the lorry is being offloaded.
3) The lorry will be positioned next to the laydown area.
4) Using the integral Hi-ab (remote control if needed) and outriggers the driver will sling and lift all items into appropriate positions in the laydown area to minimize manual handling for the tank erection crew. The tank liner will be positioned either on or as close to the tank base as possible.
5) The delivery note will be presented for signature.

Note: If the tank erection does not commence within 48 hours of delivery it will be necessary for suitable undercover dry storage to be arranged to ensure that components are not damaged or lost. This is the responsibility of the client – CST will not accept any charges for failure to comply.

5. On Site Attendance

Each operative will attend a site safety induction course given by the client. Operatives will sign a pro-forma confirming they have attended and fully understood the contents of the induction. Site will issue any permits required.
All CST method statements must be read through, signed by all CST site operatives and returned back to CST Head Office.
Each operative will sign in and out each day in the relevant diary / log book where required by the client.
Shared welfare facilities will be provided by the client.

6. First Aid

A trained first aider (supplied by client) to be on site at all times.
Emergency procedures will be to suit the site rules as indicated at the client site induction.

Rescue Procedure for confirmed space

All operatives will be confined space trained will employ the use of gas monitors & escape sets when / if relevant during the construction of the tank.

During construction of each tank ring the last panel will have the bolts left out of the last vertical seam and ½ of the top horizontal seam of the same panel. This will allow access and egress before the tank shell is jacked to allow the construction of the next ring.

At this point, verbal communication between the "outside" and "inside" operatives is maintained. Should verbal contact be lost and/or it becomes evident that the inside activities have ceased (The inside man is there only to insert tank shell bolts through the plates for the outside operative to "nut and washer"), the outside operative raises the tank shell immediately by activating the jacking equipment and the stricken operative on the internal periphery of the tank is pulled out of the tank and into clear open external space.

Emergency Services are to be contacted if required.
The nearest local NHS Accident & Emergency department is to be advised @ induction.

7. Personal Protective Equipment

All operatives will wear the following items as a minimum whilst working on site.
Hi Vis Vest
Safety Helmet
Safety Boots
Gloves
Safety Glasses
Dust Mask

Short term exposure to noise will be generated by the use of a generator/power tools.
Ear defenders are issued and all personnel are instructed to use them when carrying out tasks that generate significant noise levels.

Short term exposure to vibration will be generated by the use of power tools.
Power tools to be used for a short time & operatives to take in turns.

All work at height must be carried out using harness and safety lanyard. A working at height permit will be issued from site if required prior to any working at height being undertaken.

All manual handling is assessed at site and team lifting techniques are adopted.

8. Work Equipment

Low voltage electrical tools will be used – PAT tested.
All corded power tools are to be visually inspected prior to use.
Generator, hydraulic unit, electric drill, jigsaw, grinder, STIHL saw, concrete breaker, nut runner, tin snips and spanners.

9. Housekeeping

General waste will be cleared away and disposed of in skips provided by the client each day.

10. Working Area

Designated access routes to be agreed with the client and to be always kept clear to prevent slips, trips, and falls.

Work areas are to be barriered off to exclude non-participants when necessary.
Client to barrier off working area where required.

11. Working hours

Site working hours to be to be confirmed and agreed with client at site.

12. Client Responsibility

Client to ensure that tank & base access is always clear.

Client to ensure that the tank is drained prior to the start of the dismantle.

Client to ensure that all connections (electrical and mechanical are disconnected from the tank for the start of the dismantle.

Client is responsible for clearing the working area of all rubbish and debris prior to CST works commencing.

The waste skips are to be provided by CST Industries.

13. Dismantle of Existing Tank using Hydraulic Jacking System

1) The tank connections are to be marked on the base during dismantle to ensure that the new connections are placed in the same orientation and the GA drawing is to be updated accordingly.
2) Attach Hydraulic lifting jacks to the vertical seams in the second ring of the tank.
3) The Jacks are then coupled up to the hydraulic control unit with hydraulic pipes, the tank weight is then taken slightly with the jacks.
4) The bottom ring of sheets is cut through at base level where the tank panels meet the concrete haunch.
5) On the lowest ring, un-pin the panel fixing bolts to remove the panels, place the panels in the skip, also remove connections & ladder sections and place in the skip.
6) Once the complete ring & are removed the tank is slowly lowered down to the floor by use of the jacks.
7) The jacks are repositioned from the bottom ring and fastened to the next ring above, the weight is then taken slightly with the jacks.
8) Repeat actions until the top ring of the tank with the roof attached is down to 1 ring high, jacks will then be removed.
9) The roof flashings are unfastened and placed in the skip.
10) The BVH, float valve, platform and any other tank roof connections are unfastened and placed in the skip.
11) Each roof panel sheet is then unfastened, removed, and placed in the skip.
12) The ladder sections are removed piece by piece as the tank rings are removed and are placed in the skip.
13) The last of the tank panels are then removed and placed in the skip.

14. Tank Erection

PRELIMINARY WORK:

1) The tank bases will be checked for level and swept clean.
2) The bottom angles will be fastened to the marked position on the base with parabolts. If the tank base is out of level shims may be used to achieve level of the bottom angles.
3) The tank walls are comprised of panels. The top row of panels will be temporarily assembled to the bottom angles, thus forming the tank shape.
4) The top angles will be fastened to the top of the first ring of panels.
5) The vertical bolt seams on the top row of panels will be fully fastened and torqued up.
6) All high-level connections will be drilled into the tank's top row of panels as per the previously marked connection positions (during the tank dismantle).

ROOF ASSEMBLY:

1) Purlin Brackets will be fixed around the top angles – spaced according to the roof drawing.
2) Roof purlins will be attached to the purlin brackets across the tank.
3) A Ball Valve Housing and back plate will be fitted onto the purlins and attached in accordance with the marked infill position from the tank dismantle.
4) Roof decking will be assembled upon the purlins according to the roof drawing.
5) The roof decking will then be trimmed to follow the tanks circumference shape.
6) Roof flashing is then installed around the roof circumference.
7) The inspection Platform will be fitted into place and attached to the BVH.
8) If required a Test return Bracket will be fitted at the agreed location.

TANK ERECTION:

1) All debris is removed from the tank floor.
2) The tank liner is laid out and attached to the top angle of bolts around the perimeter of the tank.
3) All high-level fittings will be connected using protective gaskets on both sides of the liner.

JACKING:

1) Lifting jacks will be bolted to the outside of the tank at every other vertical seam and bolted down to the tank base using Parabolts.
2) The top ring of panels will be unfastened from the bottom angles.
3) The top ring c/w the roof assembly will be lifted one ring high by the jacks. This will allow the second ring of panels to be attached into position.
4) Protective poythene strips will be attached to all vertical and horizontal seams.
5) This method will continue until the tank is at full height.
6) The manway plate will be the last plate to be attached and will <u>always allow for visual contact with operatives inside the tank.</u>

ERECTION OF LADDERS:

1) Tank ladders will be attached to the tank in sections during the tank erection process.

MANWAY FITTING:

1) The manway plate will be fitted and using protective gaskets the manway body will be attached.
2) The manway lid c/w contents gauge will be fitted.

SUCTION & DRAIN FITTING:

1) The liner will be gathered away from the designated area prior to any drilling & cutting.
2) A hole will be drilled and using a 110v Jigsaw the suction aperture will be cut.
3) On completion of drilling and cutting all debris will be swept and cleared away.
4) Holes will be punched through the liner and the drain will be connected through the tank wall.
5) The suction assembly will be connected using protective gaskets.
6) The liner will be re-positioned around the perimeter of the tank.

COMPLETION:

1) 100mm to 150mm of water is to be introduced into the tank to ensure that the liner maintains the correct shape and avoids damage / failure before or during the first fill.
2) Client to be informed of completion and handover document to be signed after a visual inspection has taken place.

Name				
Signature				
Date				

Contract No: 24-T4265 – GXO Decathlon, Wellingborough, NN8 6XX

This Page must be completed & photographed then sent to the relevant Project Manager at CST listed in part 2.

Amendment Sheet

Client / Site Representative

Date:

CST Representative

Date:

CST Operatives

Name			
Signature			
Date			